LEARNING HUB MASTER SERIES • PHYSICS VOLUME 7

# Oscillations & Simple Harmonic Motion

A Rigorous Synthesis of Kinematics, Restoring Forces, Dynamic Differential Equations, Energy Conservation, Compound & Torsion Pendulums, Damped Oscillations, Driven Resonance, and Calculation Fast-Tracks

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**Theoretical Foundations:** *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Prof. R. Shankar, Yale University)

**Curriculum Synthesis:** University Physics Lecture Series & Faculty Slide Archive (Simple Harmonic Motion, Pendulums, Resonance)

**Integration & Rigor:** Complete embedding of formulas, derivations, SI units, and exam calculation shortcuts

**Visual Standard:** 4 Scalable Vector SVG Visualizations • 4 Dedicated Calculation Fast-Tracks • 4 Step-by-Step Worked Problems

Learning Hub Master Series • Pure English Master Reference Handbook • Document/Physics Directory • Published 2026

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This reference handbook synthesizes the complete mechanics of harmonic oscillatory systems into five systematic parts, progressing from fundamental kinematics and Hookean restoring dynamics to energy landscapes, damping regimes, forced resonance, and fully evaluated engineering problems.


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# Part I: Foundations of Simple Harmonic Motion

## Chapter 1: The Physics of Restoring Forces & Equilibrium

Periodic motion is motion that repeats itself at regular time intervals. Simple Harmonic Motion (SHM) represents the purest, most fundamental mathematical realization of periodic motion. Whenever any mechanical or physical system is displaced slightly from a point of stable equilibrium, it experiences a restoring force directed toward the equilibrium state.

### Hooke's Law & Universal Quadratic Potential Approximation

An ideal linear restoring force obeys Hooke's Law:

$$F_x = -kx$$

where $x$ is the displacement from equilibrium (measured in meters, $\mathrm{m}$), $k$ is the spring stiffness constant (measured in Newtons per meter, $\mathrm{N/m}$), and the negative sign indicates that the restoring force always opposes displacement.

**Taylor Expansion Rigor:** Why is Hooke's Law ubiquitous throughout classical mechanics, solid-state physics, and quantum mechanics? Consider an arbitrary smooth potential energy function $U(x)$ with a stable local minimum at $x_0 = 0$. A Taylor series expansion around equilibrium yields:

$$U(x) = U(0) + U'(0)x + \frac{1}{2}U''(0)x^2 + \frac{1}{6}U'''(0)x^3 + \dots$$

At equilibrium, the net force is zero, meaning $F(0) = -U'(0) = 0$. Shifting our energy reference so that $U(0) = 0$, and ignoring cubic and higher terms for sufficiently small displacements ($|x| \ll 1$), the potential reduces identically to:

$$U(x) \approx \frac{1}{2}\, k_{eff}\, x^2, \quad \text{where} \quad k_{eff} = \frac{d^2U}{dx^2}\Big|_{x=0} > 0$$

The corresponding force is $F(x) = -dU/dx = -k_{eff}\, x$. Thus, *any physical system slightly perturbed from a stable potential well behaves as a simple harmonic oscillator!*

### Core Kinematic Definitions

- **Displacement ($x$):** The distance and direction of the oscillating particle from its equilibrium position ($x = 0$), in meters ($\mathrm{m}$).
- **Amplitude ($A$):** The maximum magnitude of displacement from equilibrium ($A > 0$), in meters ($\mathrm{m}$).
- **Period ($T$):** The duration required to complete one full cycle of oscillation, measured in seconds ($\mathrm{s}$).
- **Frequency ($f$):** The number of complete oscillatory cycles executed per unit time, measured in Hertz ($\mathrm{Hz} = \mathrm{s}^{-1}$): $f = \frac{1}{T}$.
- **Angular Frequency ($\omega$):** The rate of change of the sinusoidal phase argument, measured in radians per second ($\mathrm{rad/s}$): $\omega = 2\pi f = \frac{2\pi}{T}$.

## Chapter 2: The Fundamental SHM Differential Equation & General Solutions

Applying Newton's Second Law ($\sum F = ma$) to a body of mass $m$ acted upon solely by a Hookean restoring force gives:

$$m\,\frac{d^2x}{dt^2} = -kx \quad \&implies; \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Defining the fundamental natural angular frequency as $\omega_0 = \sqrt{k/m}$, we arrive at the canonical second-order linear homogeneous ordinary differential equation for SHM:

$$\boldsymbol{\frac{d^2x}{dt^2} + \omega^2 x = 0 \quad \Leftrightarrow \quad \ddot{x} + \omega^2 x = 0}$$

### Analytical Solution Methods

Because this differential equation is linear and homogeneous with constant coefficients, it can be solved completely using two mathematically equivalent formalisms:

1. **Trigonometric Harmonic Formulation:** The general solution contains two independent arbitrary constants determined by initial conditions:

   $$x(t) = A\,cos(\omega t + \varphi) = C_1\,cos(\omega t) + C_2\,sin(\omega t)$$

   where $A$ is the amplitude, $\varphi$ is the initial phase angle (phase constant), $C_1 = A\,cos\,\varphi$, and $C_2 = -A\,sin\,\varphi$.
2. **Complex Exponential (Ansatz) Formulation:** Following Shankar's approach, let $x(t) = C\,e^{\alpha t}$. Substituting into $\ddot{x} + \omega^2 x = 0$ yields the characteristic algebraic equation:

   $$\alpha^2 + \omega^2 = 0 \quad \&implies; \quad \alpha = \pm\, i\omega$$

   By the principle of linear superposition, the general complex solution is $z(t) = c_1\,e^{i\omega t} + c_2\,e^{-i\omega t}$. To represent physical position on a real coordinate axis, $z(t)$ must be real ($z = z^*$), requiring $c_2 = c_1{}^*$. Expressing $c_1$ in polar form as $\frac{1}{2}A\,e^{i\varphi}$, Euler's formula $e^{\pm i\theta} = cos\,\theta \pm i\,sin\,\theta$ directly yields:

   $$x(t) = \frac{1}{2}A\,e^{i(\omega t+\varphi)} + \frac{1}{2}A\,e^{-i(\omega t+\varphi)} = A\,cos(\omega t + \varphi)$$

## Chapter 3: Kinematic Waveforms & Phase Relationships

Differentiating the displacement equation with respect to time yields the velocity and acceleration profiles. In simple harmonic motion, velocity and acceleration are also purely sinusoidal functions of the identical frequency $\omega$, but shifted systematically in phase:

**Kinematic Hierarchy of SHM**

| | | |
|---|---|---|
| **Displacement:** | $x(t) = A\,cos(\omega t + \varphi)$ | Peak magnitude: $x_{max} = A$ |
| **Velocity:** | $v(t) = \frac{dx}{dt} = -\omega A\,sin(\omega t + \varphi) = \omega A\,cos(\omega t + \varphi + \frac{\pi}{2})$ | Peak magnitude: $v_{max} = \omega A$ |
| **Acceleration:** | $a(t) = \frac{d^2x}{dt^2} = -\omega^2 A\,cos(\omega t + \varphi) = \omega^2 A\,cos(\omega t + \varphi + \pi)$ | Peak magnitude: $a_{max} = \omega^2 A$ |

**Physical Interpretation of Phase Shifts:**

- **Velocity leads displacement by $\pi/2$ radians ($90°$ or $T/4$):** When displacement is at its positive maximum ($x = +A$, turning point), the velocity is instantaneously zero ($v = 0$). As the mass rushes through the equilibrium point ($x = 0$), velocity reaches its maximum negative value ($v = -\omega A$).
- **Acceleration is in exact antiphase ($\pi$ radians or $180°$) with displacement:** Notice that $a(t) = -\omega^2 x(t)$. Acceleration is always proportional to displacement and oriented in the opposite direction. At $x = +A$, acceleration is at its most negative ($a = -\omega^2 A$); at equilibrium ($x = 0$), acceleration vanishes identically ($a = 0$).

**Figure 7.1: Phase Relationships in Simple Harmonic Motion ($x_0 = +A$, $v_0 = 0$)**

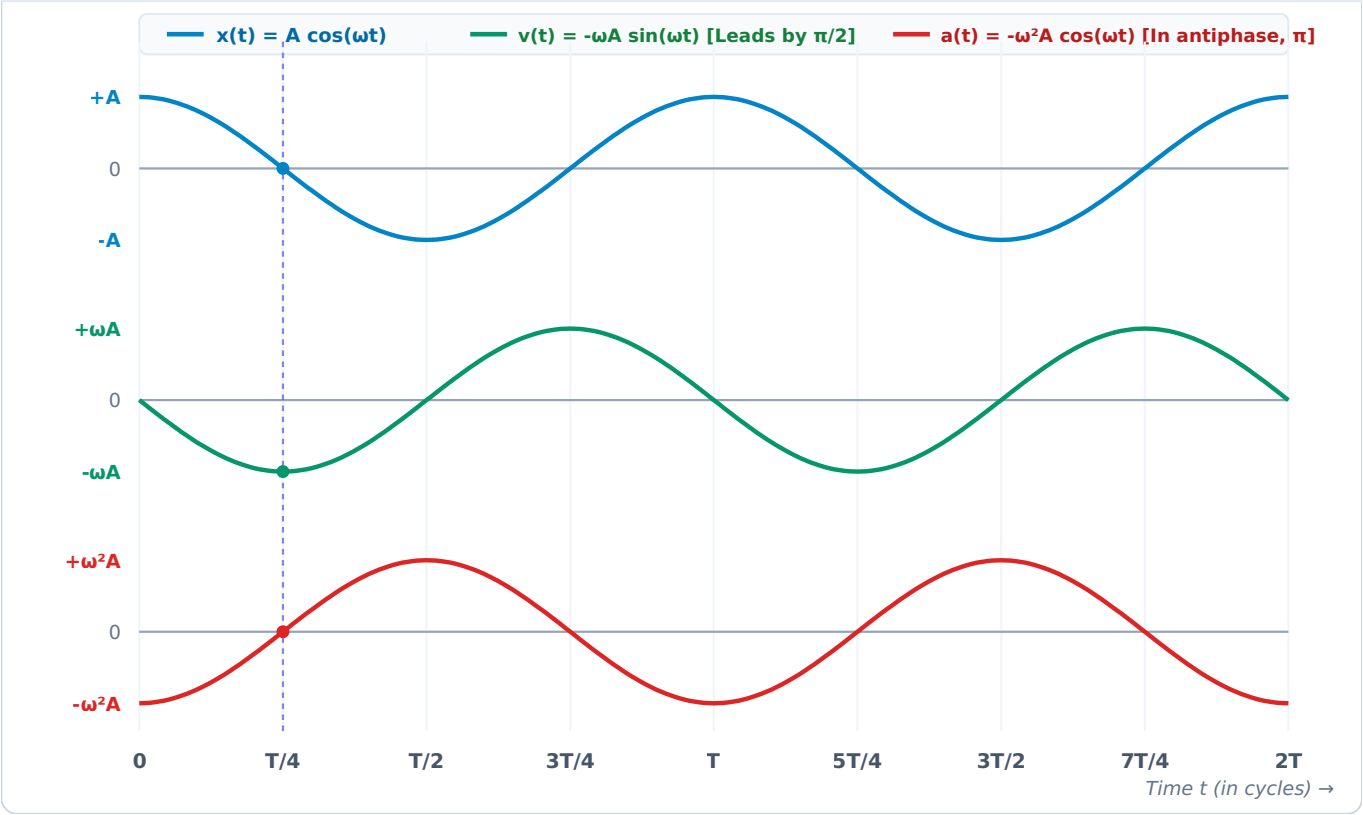



***Figure 7.1:*** *Synchronized kinematic waveforms for an undamped harmonic oscillator starting from maximum release ($x(0) = +A$, $v(0) = 0$). The vertical dashed indicator at $t = T/4$ highlights the phase progression: displacement reaches zero, velocity reaches its negative peak ($-\omega A$), and acceleration crosses zero.*

### Determination of Amplitude and Phase from Initial Conditions

Suppose an oscillator is released at $t = 0$ with initial position $x(0) = x_0$ and initial velocity $v(0) = v_0$. Evaluating the general equations at $t = 0$ gives:

$$x_0 = A\cos\varphi, \qquad v_0 = -\omega A\sin\varphi \quad \&implies; \quad \frac{-v_0}{\omega} = A\sin\varphi$$

Squaring and summing these relations eliminates $\varphi$ via $\cos^2\varphi + \sin^2\varphi = 1$, while dividing them yields $\tan\varphi$:

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2}, \qquad \tan\varphi = -\frac{v_0}{\omega x_0} \qquad (\varphi = \arctan2(-v_0/\omega, x_0))$$

## Chapter 4: Reference Circle & Superposition of Harmonic Oscillations

There is a direct geometric equivalence between **Uniform Circular Motion (UCM)** in a two-dimensional plane and **Simple Harmonic Motion (SHM)** in one dimension.

### The Reference Circle (Phasor Representation)

Imagine a reference particle $P$ revolving counterclockwise along a circle of radius $A$ in the $xy$-plane with constant angular speed $\omega$. If its angular position at $t = 0$ is $\varphi$, its angular coordinate at time $t$ is $\theta(t) = \omega t + \varphi$.

Projecting the position of particle $P$ onto the horizontal $x$-axis yields:

$$x(t) = A\cos(\omega t + \varphi)$$

The tangential velocity of $P$ is $v_{tangential} = \omega A$. Its projection onto the $x$-axis is $v_x(t) = -\omega A \sin(\omega t + \varphi)$. The centripetal acceleration of $P$ is $a_{centripetal} = \omega^2 A$ directed toward the center; its projection onto the $x$-axis is $a_x(t) = -\omega^2 A\cos(\omega t + \varphi) = -\omega^2 x(t)$.

*Conclusion: Simple harmonic motion is mathematically identical to the one-dimensional projection of uniform circular motion!*

### Superposition of Collinear Harmonic Oscillations

When two harmonic forces of the same angular frequency $\omega$ act along the same coordinate axis, the net displacement is the algebraic sum of the individual oscillations:

$$x_{net}(t) = x_1(t) + x_2(t) = A_1\cos(\omega t + \varphi_1) + A_2\cos(\omega t + \varphi_2) = A_{res}\cos(\omega t + \varphi_{res})$$

Using phasor addition (vector addition of rotating arrows of lengths $A_1$ and $A_2$ separated by phase angle $\delta = \varphi_2 - \varphi_1$):

$$A_{res} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\delta}, \qquad \tan\varphi_{res} = \frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$$

This fundamental equation appears on university physics reference sheets. If the oscillations are in phase ($\delta = 0$), $A_{res} = A_1 + A_2$ (constructive). If they are in antiphase ($\delta = \pi$), $A_{res} = |A_1 - A_2|$ (destructive).

### ⚡ FAST-TRACK 1: KINEMATIC PEAKS & PHASE DETERMINATION MENTAL SHORTCUTS

**1. Instantaneous Peak Ratios:** In any SHM problem, knowing any one peak parameter immediately fixes the others via powers of $\omega$:

$$v_{max} = \omega A, \qquad a_{max} = \omega^2 A = \omega v_{max} \quad \&implies; \quad \omega = \frac{a_{max}}{v_{max}}, \quad A = \frac{v_{max}^2}{a_{max}}$$

**2. Rapid Phase Angle Sign Rule:**

- Released from rest at positive extreme ($x_0 = +A$, $v_0 = 0$): $\varphi = 0$.
- Released from rest at negative extreme ($x_0 = -A$, $v_0 = 0$): $\varphi = \pi$.
- Passing through equilibrium moving in the $+x$ direction ($x_0 = 0$, $v_0 > 0$): $\varphi = -\pi/2$ (equivalent to $x(t) = A\ sin(\omega t)$).
- Passing through equilibrium moving in the $-x$ direction ($x_0 = 0$, $v_0 < 0$): $\varphi = +\pi/2$ (equivalent to $x(t) = -A\ sin(\omega t)$).

**3. Sine vs. Cosine Convention:** The lecture slides often express position as $x(t) = A\ sin(\omega t + \varphi')$. Since $sin(\theta + \pi/2) = cos\ \theta$, both conventions describe the exact same physical motion with $\varphi' = \varphi + \pi/2$. Never be confused on exams!

# Part II: Mechanical Systems — Springs and Pendulums

## Chapter 5: Mass-Spring Oscillators: Configurations & Gravity

The canonical mechanical oscillator comprises a block of mass $m$ connected to an ideal spring of stiffness $k$. We examine horizontal, inclined, and vertical orientations.

### The Vertical Mass-Spring System: The Gravity Invariance Theorem

A common source of confusion among physics students is whether gravity alters the frequency or period of a vertical mass-spring system. Let us prove rigorously that it does not.

#### Proof of Equilibrium Shift Invariance

When a mass $m$ is hung vertically from a spring of stiffness $k$, it stretches the spring by a static distance $\Delta y_0$ to reach static equilibrium:

$$\sum F_y = k\,\Delta y_0 - mg = 0 \quad \&implies; \quad \Delta y_0 = \frac{mg}{k}$$

Now let the mass be displaced downward by a dynamic coordinate $y$ measured from this new equilibrium position. The total extension of the spring is $\Delta y = \Delta y_0 + y$. Newton's Second Law yields:

$$m\,\frac{d^2y}{dt^2} = mg - k(\Delta y_0 + y) = mg - k\frac{mg}{k} - ky = -ky$$

$$\frac{d^2y}{dt^2} + \frac{k}{m}y = 0 \quad \&implies; \quad \omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{\Delta y_0}{g}}$$

**Core Principle:** Constant external forces (such as gravity $mg$ or an inclined plane component $mg\sin\theta$) shift the equilibrium position by $\Delta y_0 = mg/k$, but *have absolutely zero effect on the restoring force gradient or the oscillatory period!*

## Chapter 6: Spring Combinations: Series, Parallel & Spring Cutting Laws

When multiple springs are coupled to a mass, their collective behavior can be replaced by a single equivalent spring constant $k_{eff}$:

### 1. Springs in Parallel (Side-by-Side)

In a parallel configuration, all springs undergo the identical displacement $x$. The total restoring force is the sum of the individual spring forces:

$$F_{total} = -k_1x - k_2x = -(k_1 + k_2)x \quad \&implies; \quad k_{parallel} = k_1 + k_2 + \ldots + k_N$$

Parallel springs behave like electrical capacitors in parallel or resistors in series: their stiffnesses add directly, making the system **stiffer** and increasing the angular frequency $\omega$.

### 2. Springs in Series (End-to-End)

In a series configuration, massless springs transmit the identical restoring tension $F$ throughout the chain. The total extension $x_{total}$ is the sum of individual extensions:

$$x_{total} = x_1 + x_2 = \frac{|F|}{k_1} + \frac{|F|}{k_2} = |F|\frac{1}{k_{eff}} \quad \&implies; \quad \frac{1}{k_{series}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_1 + k_2}{k_1\, k_2}$$

$$k_{series} = \frac{k_1\, k_2}{k_1 + k_2} \quad \textit{(for 2 springs)}$$

Series springs behave like electrical resistors in parallel: the effective stiffness is **always less than the smallest individual stiffness**, resulting in a more compliant system and a longer period $T$.

### 3. Spring Cutting Law

An ideal spring of rest length $L_0$ and spring constant $k$ has an intrinsic material stiffness property $k \times L_0 = \text{const}$ (governed by wire shear modulus $G$ and geometry). If a spring is cut into a fraction $\eta = L'/L_0$:

$$k' = \frac{k}{\eta} = \frac{k\, L_0}{L'}$$

**Example:** Cutting a spring in half ($\eta = 1/2$) doubles its spring constant ($k' = 2k$).

### ⚡ FAST-TRACK 2: EFFECTIVE SPRING CONSTANTS & PERIOD SCALING SHORTCUTS

**1. Identical Spring Networks:** For $N$ identical springs of stiffness $k$:

$$k_{parallel} = N\,k \quad \&implies; \quad T_{parallel} = \frac{T_0}{\sqrt{N}}; \qquad k_{series} = \frac{k}{N} \quad \&implies; \quad T_{series} = \sqrt{N}\,T_0$$

**2. Mass or Stiffness Scaling Ratios:** Because $T \propto \sqrt{m/k}$:

- Quadrupling mass ($m \to 4m$) exactly doubles the period: $T \to 2T$.
- Quadrupling stiffness ($k \to 4k$) halves the period: $T \to T/2$.
- Small percentage changes ($\Delta m/m \ll 1$, $\Delta k/k \ll 1$): $\frac{\Delta T}{T} \approx \frac{1}{2}\frac{\Delta m}{m} - \frac{1}{2}\frac{\Delta k}{k}$.

**3. Two-Spring Connected Mass:** When a mass is sandwiched between two fixed walls by two springs $k_1$ and $k_2$, displacing the mass stretches one spring and compresses the other. Both forces act in the same direction! Hence, they are in **parallel**: $k_{eff} = k_1 + k_2$.

## Chapter 7: Simple Gravity Pendulum & Small-Angle Approximations

A simple pendulum consists of a point mass (bob) of mass $m$ suspended from a frictionless pivot by a light, inextensible cord of length $L$.

### Dynamical Derivation via Torque

When displaced by an angular coordinate $\theta$ from the vertical, gravity exerts a restoring torque about pivot $O$:

$$\tau = -mg\,L\,\sin\theta = I\,\frac{d^2\theta}{dt^2} = mL^2\,\frac{d^2\theta}{dt^2}$$

Dividing by $mL^2$ gives the exact, nonlinear pendulum differential equation:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$$

**The Small-Angle Approximation:** For small angular amplitudes ($\theta \ll 1\text{ rad}$, typically $\theta \le 10° \approx 0.175\text{ rad}$), the Taylor expansion $\sin\theta = \theta - \frac{\theta^3}{6} + \ldots$ permits approximating $\sin\theta \approx \theta$ with less than $0.5\%$ error. The equation reduces to the linear SHM form:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \quad \&implies; \quad \omega = \sqrt{\frac{g}{L}}, \qquad T = 2\pi\sqrt{\frac{L}{g}}$$

### Key Characteristics of the Simple Pendulum

- **Independence of Mass:** The period depends solely on length $L$ and local gravitational acceleration $g$. A heavy lead bob and a light wooden bob swing with the identical period.
- **Isochronism:** In the small-angle regime, the period is strictly independent of amplitude $\theta_0$ (Galileo's discovery).
- **Finite Amplitude Correction (Borda's Formula):** For larger release angles $\theta_0$, the period lengthens systematically:

$$T = 2\pi\sqrt{\frac{L}{g}}\left[\,1 + \frac{1}{4}\sin^2\frac{\theta_0}{2} + \frac{9}{64}\sin^4\frac{\theta_0}{2} + \dots\,\right] \approx T_0\left[\,1 + \frac{\theta_0^{\,2}}{16}\,\right]$$

## Chapter 8: Physical (Compound) Pendulum & Center of Oscillation

Any rigid body of arbitrary shape suspended from a fixed horizontal axis that does not pass through its center of mass constitutes a **physical (or compound) pendulum**.

### Dynamical Equation & Parallel Axis Theorem

Let the rigid body have total mass $M$ and moment of inertia $I_P$ about pivot axis $P$. Let $d$ denote the distance between pivot $P$ and center of mass $CM$. Displacing the body by angle $\theta$ generates a restoring torque:

$$\tau = -Mg\, d\, \sin\theta = I_P \frac{d^2\theta}{dt^2} \quad \&implies; \quad \frac{d^2\theta}{dt^2} + \frac{Mgd}{I_P}\theta = 0 \qquad (\theta \ll 1)$$

$$\omega = \sqrt{\frac{Mgd}{I_P}}, \qquad T = 2\pi\sqrt{\frac{I_P}{Mgd}}$$

By the Parallel Axis Theorem, $I_P = I_{CM} + Md^2 = M(k_g^{\,2} + d^2)$, where $k_g$ is the radius of gyration.

**Center of Oscillation ($O'$) and Equivalent Simple Pendulum:** Comparing the physical pendulum period to that of a simple pendulum $T = 2\pi\sqrt{L_{eq}/g}$ defines the equivalent simple pendulum length:

$$L_{eq} = \frac{I_P}{Md} = \frac{I_{CM} + Md^2}{Md} = d + \frac{k_g^{\,2}}{d}$$

The point $O'$ situated along the line through $P$ and $CM$ at distance $L_{eq}$ from pivot $P$ is called the **Center of Oscillation**. Remarkably, if the body is inverted and pivoted at $O'$, it swings with the *exact same period* $T$ (conjugate pivots theorem, utilized in Kater's reversible pendulum for precision $g$ measurements).

## Chapter 9: The Torsion Pendulum

A torsion pendulum consists of a rigid body suspended by a slender wire or fiber. When twisted through angle $\theta$, the wire exerts a restoring torque:

$$\tau = -\kappa\,\theta = I\frac{d^2\theta}{dt^2} \quad \&implies; \quad \frac{d^2\theta}{dt^2} + \frac{\kappa}{I}\theta = 0$$

$$\omega = \sqrt{\frac{\kappa}{I}}, \qquad T = 2\pi\sqrt{\frac{I}{\kappa}}$$

where $\kappa$ is the torsional stiffness constant of the wire (N·m/rad), determined by wire radius $r$, length $l$, and shear modulus $G$ via $\kappa = \dfrac{\pi G r^4}{2l}$.

**Figure 7.4: Comparative Geometry and Restoring Mechanisms of Pendulum Systems**

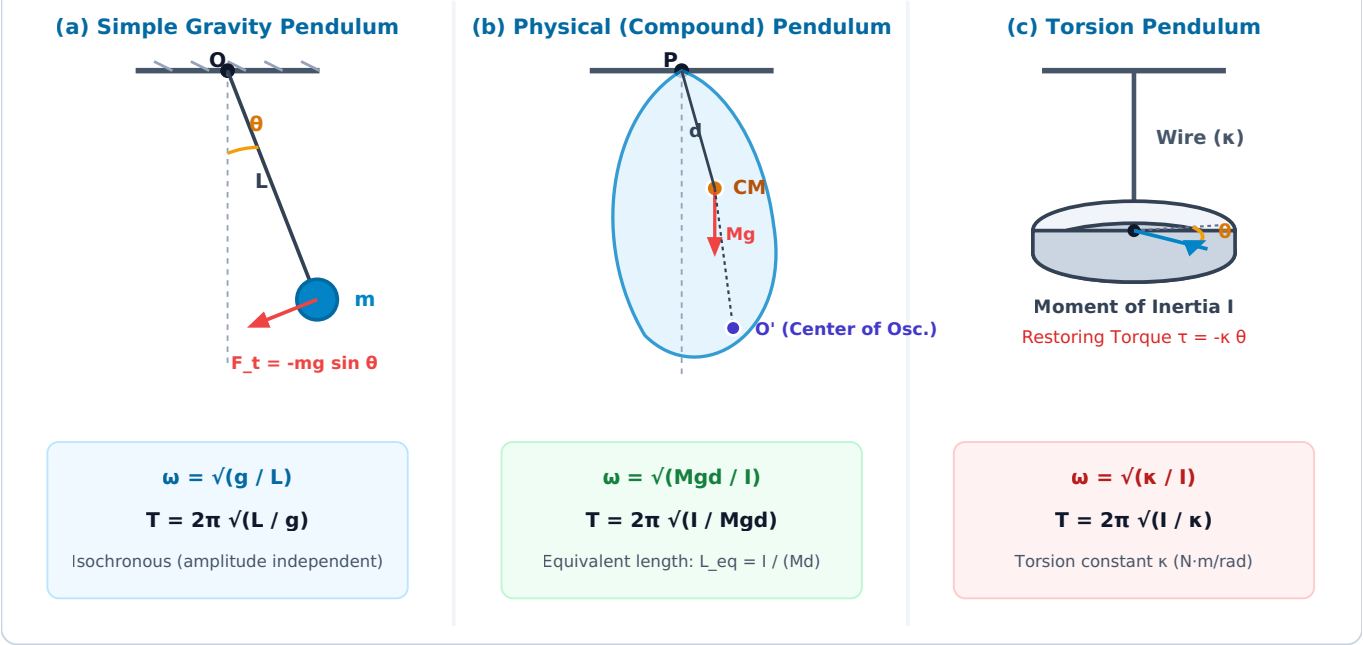



***Figure 7.4:*** *Comparative schematic architectures of oscillatory pendulum systems: (a) Simple gravity pendulum with restoring force $F_t = -mg \sin\theta$; (b) Physical pendulum showing pivot $P$, center of mass $CM$, and conjugate center of oscillation $O'$; (c) Torsion pendulum with elastic wire torque $\tau = -\kappa\theta$.*

# Part III: Energy Mechanics in Simple Harmonic Motion

## Chapter 10: Potential, Kinetic & Total Mechanical Energy Conservation

Simple harmonic motion is a conservative mechanical process. In the absence of friction and dissipative drag, energy continuously and losslessly transforms back and forth between kinetic energy stored in the moving mass and potential energy stored in the elastic restoring field.

### Instantaneous Energy Formulations

Let the position and velocity of the harmonic oscillator be $x(t) = A\ cos(\omega t + \varphi)$ and $v(t) = -\omega A\ sin(\omega t + \varphi)$.

- **Instantaneous Elastic Potential Energy ($U$):**

$$U(t) = \frac{1}{2} k\ x(t)^2 = \frac{1}{2} k\ A^2\ cos^2(\omega t + \varphi)$$

- **Instantaneous Kinetic Energy ($K$):** Recalling that $\omega^2 = k/m\ \&implies;\ m\omega^2 = k$:

$$K(t) = \frac{1}{2} m\ v(t)^2 = \frac{1}{2} m\ \omega^2\ A^2\ sin^2(\omega t + \varphi) = \frac{1}{2} k\ A^2\ sin^2(\omega t + \varphi)$$

- **Total Mechanical Energy ($E$):** Summing kinetic and potential energy yields:

$$\boldsymbol{E = K(t) + U(t) = \frac{1}{2} k\ A^2\ [\ sin^2(\omega t + \varphi) + cos^2(\omega t + \varphi)\ ] = \frac{1}{2} k\ A^2 = \frac{1}{2} m\ v_{max}{}^2 = \text{constant}}$$

**Crucial Takeaway:** Even though both $x(t)$ and $v(t)$ vary vigorously with time, the time dependence cancels completely from the total mechanical energy! Total energy is proportional to the *square of the amplitude* ($E \propto A^2$) and *square of the frequency* ($E = \frac{1}{2} m\omega^2 A^2$).

### Velocity as a Direct Function of Position $v(x)$

Solving the conservation of energy equation $\frac{1}{2} mv^2 + \frac{1}{2} kx^2 = \frac{1}{2} kA^2$ directly for speed $v$ gives:

$$v^2 = \frac{k}{m}\ (A^2 - x^2) = \omega^2\ (A^2 - x^2) \quad \&implies; \quad v(x) = \pm\ \omega\ \sqrt{A^2 - x^2}$$

This fundamental relationship appears directly on university physics examination formula sheets. It reveals:

- At the turning points ($x = \pm A$): $v = 0$, $K = 0$, and energy is purely potential: $U = E = \frac{1}{2} kA^2$.
- At the equilibrium point ($x = 0$): $U = 0$, and speed reaches its maximum: $|v| = v_{max} = \omega A$, with energy purely kinetic: $K = E = \frac{1}{2} m v_{max}{}^2$.

## Chapter 11: Energy Crossover, Equipartition & Time vs. Space Averages

Plotting $U(x) = \frac{1}{2}kx^2$ and $K(x) = \frac{1}{2}k(A^2 - x^2)$ as functions of position yields two complementary parabolas enclosed inside the constant total energy boundary line $E = \frac{1}{2}kA^2$.

**Figure 7.2: Potential, Kinetic & Total Mechanical Energy vs. Displacement x**

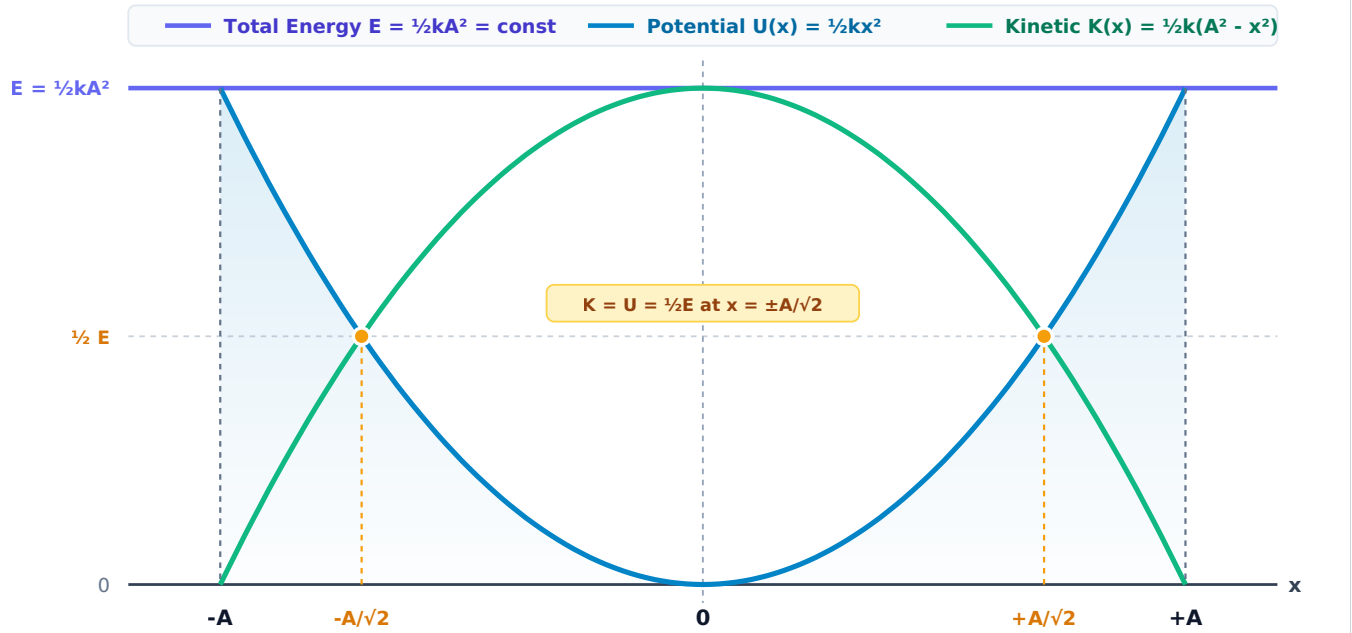



***Figure 7.2:*** *Potential energy parabola $U(x)$ (blue), kinetic energy parabola $K(x)$ (green), and total energy line $E$ (purple) as functions of displacement $x$. The gold markers at $x = \pm A/\sqrt{2}$ indicate the exact equipartition crossover locations where $K = U = \frac{1}{2}E$.*

### Equipartition Crossover Displacements

To find where kinetic energy exactly equals potential energy ($K = U$):

$$\frac{1}{2}kx^2 = \frac{1}{2}k(A^2 - x^2) \quad \&implies; \quad 2x^2 = A^2 \quad \&implies; \quad x = \pm \frac{A}{\sqrt{2}} \approx \pm\, 0.7071\, A$$

At this precise position, exactly half the total energy is kinetic and half is potential: $K = U = \frac{1}{2}E$.

### Time-Averaged vs. Space-Averaged Energies

A classic conceptual problem in university physics is the distinction between the time-averaged energy and the space-averaged energy:

- **Time Average Over One Full Period ($T$):** Because $\langle\, cos^2(\omega t)\,\rangle_{time} = \langle\, sin^2(\omega t)\,\rangle_{time} = \frac{1}{2}$ over a cycle:

  $$\langle\, U\,\rangle_{time} = \frac{1}{T}\int_0^T U(t)\, dt = \frac{1}{4}kA^2 = \frac{1}{2}E, \qquad \langle\, K\,\rangle_{time} = \frac{1}{4}kA^2 = \frac{1}{2}E$$

  Over time, energy is shared equally: $\langle\, K\,\rangle_{time} = \langle\, U\,\rangle_{time} = \frac{1}{2}E$.

- **Space Average Across the Interval $[-A, +A]$:** Integrating across spatial position gives:

$$\langle U \rangle_{space} = \frac{1}{2A}\int_{-A}^{+A} \frac{1}{2}kx^2\,dx = \frac{1}{6}kA^2 = \frac{1}{3}E, \qquad \langle K \rangle_{space} = \frac{2}{3}E$$

Why is the space average of $U$ smaller than the time average? Because the oscillator moves slowly near the turning points ($|x| \approx A$) and rapidly through the center ($x \approx 0$), it spends much more *time* at large displacements than at small displacements!

### ⚡ FAST-TRACK 3: EQUIPARTITION & ENERGY FRACTION SHORTCUTS AT SPECIAL POINTS

| Position ($x$) | Potential Energy ($U/E$) | Kinetic Energy ($K/E$) | Speed ($v/v_{max}$) | Mental Calculation Shortcut |
|---|---|---|---|---|
| $x = 0$ | $0$ (0%) | $1$ (100%) | $1.000$ | Maximum velocity transit |
| $x = \pm\frac{1}{2}A$ | $\frac{1}{4}$ (25%) | $\frac{3}{4}$ (75%) | $\frac{\sqrt{3}}{2} \approx 0.866$ | Halfway to turning point, still retaining 75% KE! |
| $x = \pm\frac{1}{\sqrt{2}}A$ | $\frac{1}{2}$ (50%) | $\frac{1}{2}$ (50%) | $\frac{1}{\sqrt{2}} \approx 0.707$ | Equipartition crossover point ($K = U$) |
| $x = \pm\frac{\sqrt{3}}{2}A$ | $\frac{3}{4}$ (75%) | $\frac{1}{4}$ (25%) | $\frac{1}{2} = 0.500$ | Speed drops to exactly half of $v_{max}$ |
| $x = \pm A$ | $1$ (100%) | $0$ (0%) | $0.000$ | Turning point, instantaneous rest |

# Part IV: Damped Oscillations and Driven Resonance

## Chapter 12: Damped Harmonic Oscillations: The Three Regimes

Real-world mechanical systems invariably dissipate energy to their environment through friction, air resistance, or internal material hysteresis. For laminar flow and modest velocities, the dissipative drag force is modeled accurately as linearly proportional and opposite to velocity:

$$F_{drag} = -b\,v = -b\,\frac{dx}{dt}$$

where $b$ is the viscous damping coefficient (measured in $\mathrm{N{\cdot}s/m} = \mathrm{kg/s}$). Applying Newton's Second Law:

$$m\,\frac{d^2x}{dt^2} = -kx - b\,\frac{dx}{dt} \quad \&implies; \quad m\,\frac{d^2x}{dt^2} + b\,\frac{dx}{dt} + kx = 0$$

Dividing by $m$ and defining the damping parameter $\gamma = \frac{b}{2m}$ and natural frequency $\omega_0 = \sqrt{k/m}$ yields the canonical damped differential equation:

$$\boldsymbol{\frac{d^2x}{dt^2} + 2\gamma\,\frac{dx}{dt} + \omega_0{}^2 x = 0}$$

Substituting the exponential ansatz $x(t) = C\,e^{\lambda t}$ gives the characteristic quadratic equation:

$$\lambda^2 + 2\gamma\lambda + \omega_0{}^2 = 0 \quad \&implies; \quad \lambda_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0{}^2} = -\frac{b}{2m} \pm \sqrt{\frac{b^2}{4m^2} - \frac{k}{m}}$$

**Figure 7.3: Comparison of Damped Oscillation Regimes & Exponential Envelope Decay**

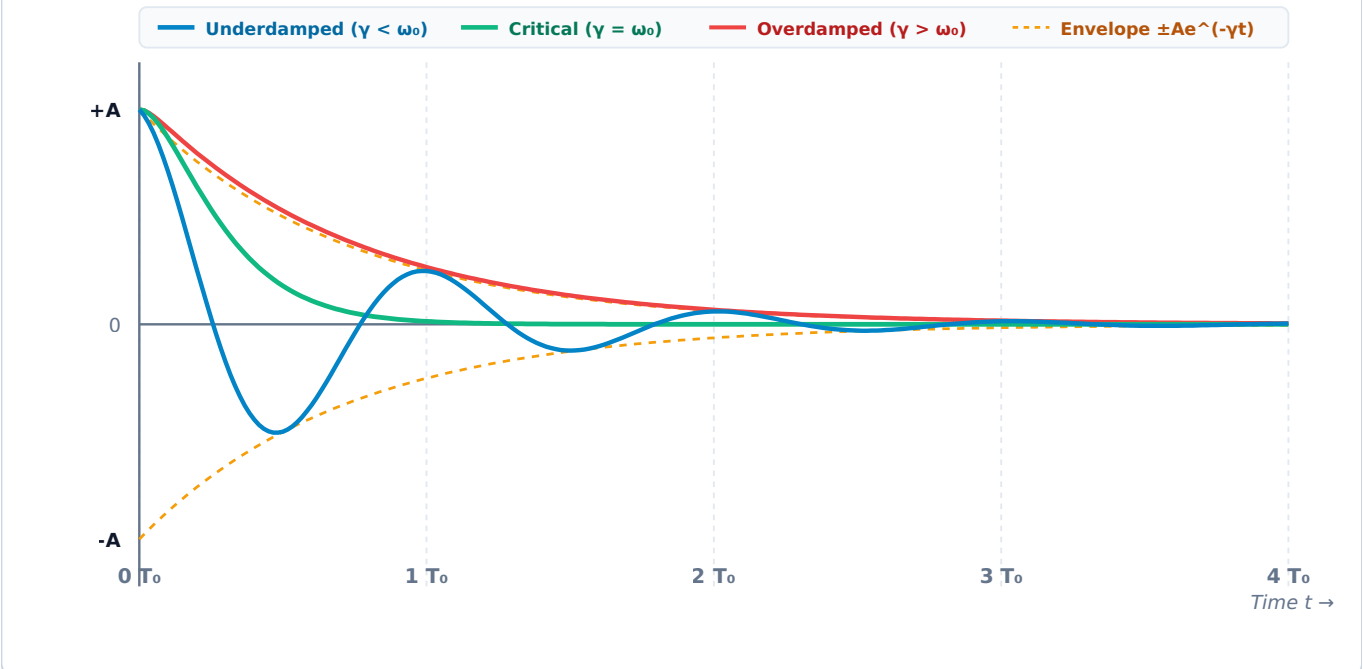



***Figure 7.3:*** *Dynamic displacement trajectories across the three damping regimes: Underdamped (blue curve oscillating within exponential decay envelope $\pm Ae^{-\gamma t}$), Critically Damped (green curve returning to equilibrium in minimal time without oscillation), and Overdamped (red curve experiencing sluggish non-oscillatory asymptotic relaxation).*

### Classification of Damping Regimes

1. **Underdamped ($\gamma < \omega_0$ or $b^2 < 4mk$):** The roots are complex conjugates: $\lambda_{1,2} = -\gamma \pm i\omega_d$, where the damped angular frequency is:

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} < \omega_0$$

The motion is an oscillation whose amplitude decays exponentially:

$$x(t) = A_0\, e^{-\gamma t}\cos(\omega_d t + \varphi) = A_0\, e^{-(b/2m)t}\cos(\omega_d t + \varphi)$$

The pseudoperiod is lengthened: $T_d = 2\pi/\omega_d > T_0$.

2. **Critically Damped ($\gamma = \omega_0$ or $b^2 = 4mk$):** The roots coalesce into a repeated real root $\lambda = -\gamma$. The general solution is:

$$x(t) = (C_1 + C_2 t)\, e^{-\gamma t}, \quad \text{with} \quad C_1 = x(0), \quad C_2 = v(0) + \gamma x(0)$$

The system relaxes back to equilibrium in the *minimum possible time without oscillating*. Essential for vehicle shock absorbers and sensitive measuring apparatus.

3. **Overdamped ($\gamma > \omega_0$ or $b^2 > 4mk$):** Both roots are purely real, distinct, and negative ($\lambda_1, \lambda_2 < 0$):

$$x(t) = C_1\, e^{\lambda_1 t} + C_2\, e^{\lambda_2 t}, \quad \text{where} \quad \lambda_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$$

The system does not oscillate; it creeps back to equilibrium very slowly, dominated at long times by the gentler decay rate $\lambda_1 = -\gamma + \sqrt{\gamma^2 - \omega_0^2}$.

# Chapter 13: Driven / Forced Oscillations, Phase Lag & Steady-State Amplitude

If an external time-periodic driving force $F_{ext}(t) = F_0\ cos(\omega t)$ is applied to a damped harmonic oscillator, the equation of motion becomes:

$$m\ \frac{d^2x}{dt^2} + b\ \frac{dx}{dt} + kx = F_0\ cos(\omega t)$$

Dividing through by mass $m$:

$$\frac{d^2x}{dt^2} + 2\gamma\ \frac{dx}{dt} + {\omega_0}^2 x = \frac{F_0}{m}\ cos(\omega t)$$

The complete general solution is the sum of a transient solution $x_c(t)$ (solution to the homogeneous equation, which decays to zero via $e^{-\gamma t}$) and a steady-state particular solution $x_p(t)$ oscillating at the driving frequency $\omega$:

$$x(t) = x_{transient}(t) + x_{steady\text{-}state}(t) \quad \longrightarrow_{t \gg 1/\gamma} \quad x_{steady\text{-}state}(t) = A(\omega)\ cos(\omega t - \varphi)$$

### Steady-State Resonance Equations

Solving by complex impedance or phasor analysis yields the authoritative formulas:

$$A(\omega) = \frac{F_0 / m}{\sqrt{({\omega_0}^2 - \omega^2)^2 + 4\gamma^2\omega^2}} = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + b^2\omega^2}}$$

$$tan\ \varphi = \frac{2\gamma\omega}{{\omega_0}^2 - \omega^2} = \frac{b\omega}{k - m\omega^2}, \qquad (0 \le \varphi \le \pi)$$

### Phase Lag Evolution & Resonance Phenomena

The phase angle $\varphi$ represents the lag of the mass displacement behind the driving force:

- **Low Frequency Regime ($\omega \ll \omega_0$):** $\varphi \to 0$. The oscillator moves almost perfectly in phase with the driving force. The response is stiffness-dominated: $A(\omega) \approx F_0/k$.
- **Resonance Condition ($\omega = \omega_0$):** $\varphi = \pi/2 = 90°$. The displacement lags the force by $90°$. Because velocity leads displacement by $90°$, *the velocity is in perfect phase with the driving force!* The force continuously delivers positive mechanical work:

  $$P(t) = F_{ext}(t) \cdot v(t) = F_0\ cos(\omega_0 t) \times v_{max}\ cos(\omega_0 t) = F_0\ v_{max}\ cos^2(\omega_0 t) \ge 0$$

  Rate of energy transfer from driver to oscillator is maximized.
- **High Frequency Regime ($\omega \gg \omega_0$):** $\varphi \to \pi = 180°$. The displacement moves in exact opposition to the force. The motion is inertia-dominated: $A(\omega) \approx F_0 / (m\omega^2)$.

### Exact Resonance Frequencies

Setting $dA/d\omega = 0$ reveals that the displacement amplitude peaks at a frequency slightly below $\omega_0$:

$$\omega_{resonance,\ displacement} = \sqrt{\omega_0^2 - 2\gamma^2} = \sqrt{\frac{k}{m} - \frac{b^2}{2m^2}} \quad (\text{valid for } \gamma < \omega_0/\sqrt{2})$$

By contrast, the **velocity amplitude** ($v_{max}(\omega) = \omega\, A(\omega)$) and the **average power absorption** peak precisely at the natural frequency: $\omega_{resonance,\ velocity} = \omega_0$.

# Chapter 14: Quality Factor Q, Resonance Bandwidth & Dissipation per Cycle

The **Quality Factor ($Q$)** is a dimensionless parameter that characterizes the sharpness of the resonance peak and the degree of energy retention in an oscillatory system.

### Definitions & Physical Formulations of Q

$$Q = \frac{\omega_0}{2\gamma} = \frac{m\omega_0}{b} = \frac{\sqrt{mk}}{b} = \frac{\omega_0}{\Delta\omega}$$

**Energy Interpretation:** For weakly damped systems ($Q \gg 1$), $Q$ represents $2\pi$ times the ratio of total stored mechanical energy to the energy dissipated per single cycle:

$$Q = 2\pi \frac{E}{|\Delta E_{cycle}|} \quad &implies; \quad \frac{|\Delta E_{cycle}|}{E} \approx \frac{2\pi}{Q}$$

### Resonance Bandwidth & Half-Power Points

The resonance curve has a full width at half-maximum (FWHM) of power absorption, denoted by $\Delta\omega = \omega_2 - \omega_1$, where power drops to half its peak value ($P = \frac{1}{2}P_{max}$):

$$\Delta\omega \approx 2\gamma = \frac{b}{m} = \frac{\omega_0}{Q} \quad &implies; \quad \Delta f = \frac{f_0}{Q}$$

- **High $Q$ (light damping, $Q > 100$):** Tall, exceptionally narrow resonance peak. Outstanding frequency selectivity (e.g., radio receivers, quartz clock crystals, atomic clocks).
- **Low $Q$ (heavy damping, $Q < 5$):** Broad, blunt resonance peak with rapid energy dissipation (e.g., suspension systems, seismographs, acoustic damping baffles).

### ⚡ FAST-TRACK 4: QUALITY FACTOR, BANDWIDTH & ENERGY DECAY SHORTCUTS

**1. Rate of Energy Decay:** Because energy $E \propto A^2$ and amplitude decays as $A(t) = A_0\, e^{-\gamma t}$, total energy decays at twice the rate:

$$E(t) = E_0\, e^{-2\gamma t} = E_0\, e^{-(b/m)t} = E_0\, e^{-t/\tau_E}$$

where the **energy relaxation time constant** is $\tau_E = \frac{1}{2\gamma} = \frac{m}{b} = \frac{Q}{\omega_0}$.

**2. Energy Half-Life ($t_{1/2}$):** The time required for mechanical energy to drop to 50% of its initial value is:

$$t_{1/2} = \tau_E \ln 2 = \frac{m \ln 2}{b} \approx 0.693\, \frac{m}{b}$$

**3. Peak Resonance Amplitude Shortcut:** At resonance ($\omega = \omega_0$), the steady-state amplitude simplifies directly to:

$$A(\omega_0) = \frac{F_0}{b\, \omega_0} = \frac{F_0}{k} \times Q = A_{static} \times Q$$

The Quality Factor $Q$ is the direct dynamic magnification factor over the static deflection!

# Part V: Comprehensive Step-by-Step Worked Problems

## Worked Problem 1: Mass-Spring Kinematics & Arbitrary Initial Phase Matching

KINEMATICS & ENERGY

**Problem Statement:** A block of mass $m = 0.500\text{ kg}$ is attached to a horizontal ideal spring of stiffness $k = 78.96\text{ N/m}$ on a frictionless track. At time $t = 0$, the block is displaced to $x_0 = +0.150\text{ m}$ and launched with an initial velocity $v_0 = -1.20\text{ m/s}$. Determine: (a) angular frequency $\omega$, frequency $f$, and period $T$; (b) amplitude $A$ and phase constant $\varphi$; (c) the explicit equation of motion $x(t)$; and (d) maximum speed $v_{max}$, maximum acceleration $a_{max}$, and total mechanical energy $E$.

### 1. Given Parameters:

$m = 0.500\text{ kg}, \quad k = 78.96\text{ N/m}, \quad x_0 = +0.150\text{ m}, \quad v_0 = -1.20\text{ m/s}.$

### 2. Symbolic Derivations & Numerical Evaluations:

(a) Angular frequency, frequency, and period:

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{78.96}{0.500}} = \sqrt{157.92} = 12.566\text{ rad/s} \approx 4\pi\text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{12.566}{6.283} = 2.00\text{ Hz}, \qquad T = \frac{1}{f} = 0.500\text{ s}$$

(b) Amplitude $A$ and phase angle $\varphi$:

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} = \sqrt{(0.150)^2 + \left(\frac{-1.20}{12.566}\right)^2} = \sqrt{0.0225 + 0.00912} = 0.1778\text{ m} = 17.8\text{ cm}$$

$$\tan\varphi = -\frac{v_0}{\omega x_0} = -\frac{-1.20}{(12.566)(0.150)} = +\frac{1.20}{1.885} = +0.6366 \quad \&\text{implies;} \quad \varphi = \arctan(0.6366) = +0.567\text{ rad} \approx +32.5°$$

(c) Displacement equation: $x(t) = 0.178\cos(12.57\,t + 0.567)\text{ m}$.

(d) Maximum kinematic values and total energy:

$$v_{max} = \omega A = (12.566)(0.1778) = 2.234\text{ m/s}, \qquad a_{max} = \omega^2 A = (157.92)(0.1778) = 28.08\text{ m/s}^2$$

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(78.96)(0.1778)^2 = 1.248\text{ J}$$

**✓ Physical Sanity & Boundary Check:**

Compute total energy directly from initial values: $E_0 = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2 = \frac{1}{2}(0.500)(-1.20)^2 + \frac{1}{2}(78.96)(0.150)^2 = 0.360\text{ J} + 0.8883\text{ J} = 1.2483\text{ J}$. Exact match to $\frac{1}{2}kA^2$! Also, because $x_0 > 0$ and $v_0 < 0$, the mass is moving toward equilibrium, verifying $0 < \varphi < \pi/2$.

## Worked Problem 2: Vertical Spring Suspension: Parallel vs. Series Configurations

SPRING SYSTEMS

**Problem Statement:** A heavy instrumentation package of mass $m = 1.50\text{ kg}$ is suspended vertically using two identical springs, each of unstretched length $L_0 = 0.200\text{ m}$ and spring constant $k = 300\text{ N/m}$. Calculate: (a) the equivalent spring constant $k_{eff}$, static equilibrium stretch $\Delta y_0$, and oscillation period $T_p$ when connected in **parallel**; (b) the corresponding values $k_{eff}$, $\Delta y_0$, and $T_s$ when connected in **series**; and (c) the exact ratio $T_{series} / T_{parallel}$. (Use $g = 9.80\text{ m/s}^2$).

### 1. Parallel Connection Analysis:

$$k_{parallel} = k_1 + k_2 = 300 + 300 = 600\text{ N/m}$$

$$\Delta y_{0,par} = \frac{mg}{k_{parallel}} = \frac{(1.50)(9.80)}{600} = \frac{14.70}{600} = 0.0245\text{ m} = 2.45\text{ cm}$$

$$T_p = 2\pi\sqrt{\frac{m}{k_{parallel}}} = 2\pi\sqrt{\frac{1.50}{600}} = 2\pi\sqrt{0.0025} = 2\pi(0.050) = 0.3142\text{ s} \approx \frac{\pi}{10}\text{ s}$$

### 2. Series Connection Analysis:

$$k_{series} = \frac{k \times k}{k + k} = \frac{k}{2} = \frac{300}{2} = 150\text{ N/m}$$

$$\Delta y_{0,ser} = \frac{mg}{k_{series}} = \frac{14.70}{150} = 0.0980\text{ m} = 9.80\text{ cm} \quad (\text{exactly } 4 \times \Delta y_{0,par})$$

$$T_s = 2\pi\sqrt{\frac{m}{k_{series}}} = 2\pi\sqrt{\frac{1.50}{150}} = 2\pi\sqrt{0.0100} = 2\pi(0.100) = 0.6283\text{ s} \approx \frac{\pi}{5}\text{ s}$$

### 3. Ratio of Oscillation Periods:

$$\frac{T_{series}}{T_{parallel}} = \sqrt{\frac{k_{parallel}}{k_{series}}} = \sqrt{\frac{600}{150}} = \sqrt{4} = 2.00$$

✓ **Physical Sanity & Boundary Check:**
Notice that $T = 2\pi\sqrt{\Delta y_0/g}$. For parallel: $2\pi\sqrt{0.0245/9.80} = 2\pi\sqrt{0.0025} = 0.314\text{ s}$. For series: $2\pi\sqrt{0.0980/9.80} = 2\pi\sqrt{0.0100} = 0.628\text{ s}$. Perfect internal consistency. Connecting springs in series softens the system by a factor of 4 relative to parallel, exactly doubling the period.

## Worked Problem 3: Physical Pendulum (Pivoted Rod) & Conjugate Center of Oscillation

PENDULUM DYNAMICS

**Problem Statement:** A slender, uniform metallic rod of mass $M = 1.20\text{ kg}$ and length $L = 0.800\text{ m}$ is suspended from a frictionless pivot located at one extreme end. (a) Derive the moment of inertia $I_P$ about the pivot. (b) Calculate the angular frequency $\omega$ and period $T$ of small oscillations. (c) Determine the length $L_{eq}$ of an equivalent simple pendulum and locate the Center of Oscillation $O'$. (d) Calculate the period if the rod is inverted and pivoted at $O'$.

### 1. Symbolic Derivations:

(a) Moment of inertia about end pivot: $I_P = \frac{1}{3}ML^2$.

(b) Distance from pivot to center of mass is $d = L/2$. Applying the physical pendulum equation:

$$T = 2\pi\sqrt{\frac{I_P}{Mgd}} = 2\pi\sqrt{\frac{\frac{1}{3}ML^2}{Mg(L/2)}} = 2\pi\sqrt{\frac{2L}{3g}}$$

(c) Equivalent simple pendulum length:

$$L_{eq} = \frac{I_P}{Md} = \frac{\frac{1}{3}ML^2}{M(L/2)} = \frac{2}{3}L$$

### 2. Numerical Evaluations:

$$T = 2\pi\sqrt{\frac{2(0.800)}{3(9.80)}} = 2\pi\sqrt{\frac{1.60}{29.40}} = 2\pi\sqrt{0.05442} = 2\pi(0.2333) = 1.466\text{ s}$$

$$\omega = \frac{2\pi}{T} = 4.286\text{ rad/s}, \qquad L_{eq} = \frac{2}{3}(0.800\text{ m}) = 0.5333\text{ m} = 53.3\text{ cm}$$

(d) If pivoted at $O'$, the new pivot distance to CM is $d' = L_{eq} - d = \frac{2}{3}L - \frac{1}{2}L = \frac{1}{6}L$. By Parallel Axis Theorem, $I_{O'} = \frac{1}{12}ML^2 + M(\frac{1}{6}L)^2 = \frac{1}{9}ML^2$. The new period is:

$$T' = 2\pi\sqrt{\frac{\frac{1}{9}ML^2}{Mg(\frac{1}{6}L)}} = 2\pi\sqrt{\frac{6}{9}\frac{L}{g}} = 2\pi\sqrt{\frac{2L}{3g}} = 1.466\text{ s}$$

**✓ Physical Sanity & Boundary Check:**
A simple pendulum of length $L = 0.800\text{ m}$ has period $T_{simple} = 2\pi\sqrt{0.800/9.80} = 1.795\text{ s}$. The uniform rod period is $T_{phys} = \sqrt{2/3}\ T_{simple} \approx 0.816 \times 1.795 = 1.466\text{ s}$. Because mass is distributed throughout the rod rather than concentrated at the extreme end, its effective pendulum length is only $2/3\ L$, making it swing faster.

## Worked Problem 4: Damped Oscillator Dynamics, Quality Factor Q & Energy Decay Time

DAMPING & DISSIPATION

**Problem Statement:** An underdamped oscillator consists of a mass $m = 0.250\text{ kg}$, spring constant $k = 100.0\text{ N/m}$, and linear damping coefficient $b = 0.050\text{ kg/s}$. Determine: (a) natural frequency $\omega_0$ and damping coefficient $\gamma$; (b) damped angular frequency $\omega_d$ and percentage frequency shift; (c) Quality factor $Q$; (d) the time required for total mechanical energy to decay to $1.0\%$ of its initial value; and (e) the fraction of energy lost during the first complete cycle.

### 1. Frequencies and Damping Parameters:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{100.0}{0.250}} = \sqrt{400} = 20.00\text{ rad/s}, \qquad \gamma = \frac{b}{2m} = \frac{0.050}{2(0.250)} = 0.100\text{ s}^{-1}$$

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{400 - (0.100)^2} = \sqrt{399.99} = 19.99975\text{ rad/s}$$

The fractional frequency shift is $\frac{\omega_0 - \omega_d}{\omega_0} \approx \frac{\gamma^2}{2\omega_0^2} = \frac{0.010}{800} = 0.0000125 = 0.00125\%$ (negligible).

### 2. Quality Factor Q:

$$Q = \frac{\omega_0}{2\gamma} = \frac{20.00}{2(0.100)} = \frac{20.00}{0.200} = 100.0$$

### 3. Energy Decay to 1.0%:

Total mechanical energy decays as $E(t) = E_0\, e^{-2\gamma t}$. Setting $E(t)/E_0 = 0.010$:

$$e^{-2\gamma t} = 0.010 \quad \&implies; \quad -2\gamma t = \ln(0.010) = -4.605 \quad \&implies; \quad t = \frac{4.605}{2(0.100)} = \frac{4.605}{0.200} = 23.03\text{ s}$$

### 4. Energy Lost in the First Cycle:

The period of oscillation is $T_d = \frac{2\pi}{\omega_d} \approx \frac{2\pi}{20.00} = 0.31416\text{ s}$.

$$\frac{\Delta E_{cycle}}{E_0} = 1 - e^{-2\gamma T_d} = 1 - e^{-2(0.100)(0.31416)} = 1 - e^{-0.06283} \approx 1 - (1 - 0.06283) = 0.0609 = 6.09\%$$

**✓ Physical Sanity & Boundary Check:**

Compare with the Fast-Track approximation $\frac{\Delta E}{E} \approx \frac{2\pi}{Q} = \frac{2\pi}{100} = 0.06283 = 6.28\%$. The exact exponential result is $6.09\%$. The approximation holds with outstanding accuracy ($\sim 3\%$ relative difference) confirming the high-$Q$ regime.

# Appendix: Master Reference Formula Sheet & Summary

Consolidated physical constants, canonical differential equations, operational solutions, and key calculation rules for university examinations.

| Oscillatory System | Governing Differential Equation | Natural Frequency & Period | Key Parameters |
|---|---|---|---|
| **Simple Harmonic Motion (1D)** | $\ddot{x} + \omega^2 x = 0$ | $\omega = 2\pi f = \frac{2\pi}{T}$ | $x(t) = A\cos(\omega t + \varphi)$ |
| **Horizontal Mass-Spring** | $m\ddot{x} + kx = 0$ | $\omega = \sqrt{k/m}$, $T = 2\pi\sqrt{m/k}$ | $E = \frac{1}{2}kA^2 = \frac{1}{2}mv_{max}^2$ |
| **Vertical Mass-Spring** | $m\ddot{y} + ky = 0$ | $\omega = \sqrt{g/\Delta y_0}$, $T = 2\pi\sqrt{\Delta y_0/g}$ | Equilibrium shift: $\Delta y_0 = mg/k$ |
| **Springs in Parallel** | $k_{eff} = k_1 + k_2 + \ldots + k_N$ | $\omega = \sqrt{(\sum k_i)/m}$ | Stiffer; displacements equal |
| **Springs in Series** | $\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_1+k_2}{k_1 k_2}$ | $\omega = \sqrt{k_{eff}/m}$ | More compliant; tension equal |
| **Simple Gravity Pendulum** | $\theta + \frac{g}{L}\theta = 0$ $(\theta \ll 1)$ | $\omega = \sqrt{g/L}$, $T = 2\pi\sqrt{L/g}$ | Independent of bob mass $m$ |
| **Physical (Compound) Pendulum** | $\theta + \frac{Mgd}{I_P}\theta = 0$ | $\omega = \sqrt{\frac{Mgd}{I_P}}$, $T = 2\pi\sqrt{\frac{I_P}{Mgd}}$ | Equiv. length: $L_{eq} = \frac{I_P}{Md}$ |
| **Torsion Pendulum** | $\theta + \frac{\kappa}{I}\theta = 0$ | $\omega = \sqrt{\kappa/I}$, $T = 2\pi\sqrt{I/\kappa}$ | Restoring torque: $\tau = -\kappa\theta$ |
| **Underdamped Motion ($\gamma < \omega_0$)** | $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$ | $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ | Decay envelope: $\pm Ae^{-\gamma t}$ |
| **Critically Damped ($\gamma = \omega_0$)** | $b^2 = 4mk$ | No oscillation | $x(t) = (C_1 + C_2 t)e^{-\gamma t}$ |
| **Overdamped ($\gamma > \omega_0$)** | $b^2 > 4mk$ | No oscillation | $x(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$ |
| **Driven Resonance Amplitude** | $m\ddot{x} + b\dot{x} + kx = F_0\cos(\omega t)$ | $\omega_{res} = \sqrt{\omega_0^2 - 2\gamma^2}$ | $A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2-\omega^2)^2 + 4\gamma^2\omega^2}}$ |
| **Quality Factor & Bandwidth** | $Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{2\gamma} = \frac{\sqrt{mk}}{b}$ | $\Delta\omega = 2\gamma = \frac{b}{m}$ | Cycle energy loss: $\frac{\Delta E}{E} \approx \frac{2\pi}{Q}$ |

### Common Pitfalls to Avoid on Examinations

- **Confusing $\omega$, $f$, and $T$:** Always check units! $\omega$ is in rad/s, $f$ is in Hz (rev/s), and $T$ is in s. Multiplying or dividing by $2\pi$ is the most frequent calculation error.
- **Phase Sign Convention:** In $x(t) = A\cos(\omega t + \varphi)$, if $v_0 < 0$ when $x_0 > 0$, $\varphi$ is positive. Remember $\tan\varphi = -v_0 / (\omega x_0)$.

- **Vertical Gravity Trap:** Gravity changes the static equilibrium position $\Delta y_0 = mg/k$, but *never* changes the period $T = 2\pi\sqrt{m/k}$.
- **Series vs. Parallel Springs:** Parallel adds directly ($k_1 + k_2$); series adds reciprocally ($1/k_1 + 1/k_2$). Connecting springs in series always makes the system *more flexible*, increasing the period.
- **Damped Frequency vs. Natural Frequency:** In underdamped motion, the frequency $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ is *strictly smaller* than the undamped frequency $\omega_0$, and the period is strictly longer.